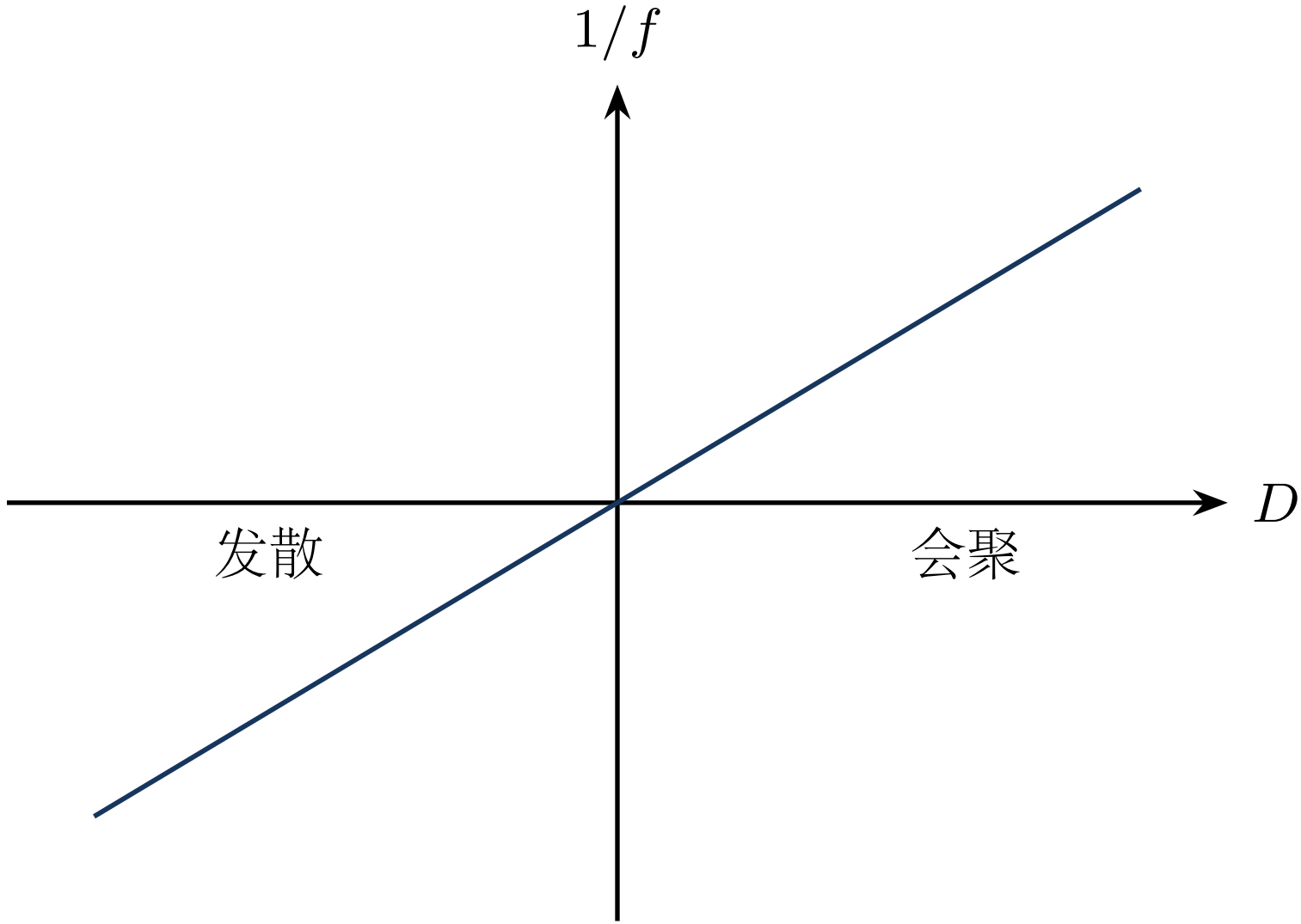

$$D = 2p_A - p_L - p_R$$

$$1/f = (n_A - n_B)D/(n_B\sigma)$$